

Crab Pulsar Observations Using the 42ft Telescope at the Jodrell Bank Observatory

Third Year Lab Report

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(Experiment performed in collaboration with Finley Groom)

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The objective of this experiment was to observe the Crab Pulsar (PSR B0531+21) with the 42-ft telescope at the Jodrell Bank Observatory to determine the period of its rotation. After applying corrections including the Dispersion Measure, the rotation periods measured on 11 November 2025 and 19 November 2025 were found to be $(0.0338482160 \pm 1 \times 10^{-10})$ s and $(0.0338486878 \pm 2 \times 10^{-10})$ s, respectively. These results were compared to established reference values and were determined to be inconsistent.

1. INTRODUCTION

Since the inception of radio astronomy in the early 20th century, pulsars have been of great interest to astronomers because of their extreme cosmic conditions [1]. Pulsars, a type of neutron star, are the collapsed cores of massive stars that spin rapidly and regularly, and possess extremely strong magnetic fields due to a combination of factors: the compression of the star's original magnetic field to a small radius and the effects of internal quantum processes. They emit beams of radiation because of their misaligned magnetic fields that accelerate charged particles to the poles, and this can be observed on Earth [2, p. 60]. The beams are detected with radio telescopes, which are used to perform sensitive observations to obtain the periodicity of the pulsar. In this experiment, the 42-ft telescope at the Jodrell Bank Observatory was used on two separate days, one week apart, to determine the period of the Crab Pulsar and to calculate parameters such as its magnetic field strength and characteristic age.

2. THEORY AND METHODOLOGY

2.1. Obtaining the Dispersion Measure

As the radio-frequency beams of the pulsar traverse the interstellar medium, it experiences a smearing effect which causes a frequency dependent delay in the arrival time of the pulse. To reverse this effect, the dispersion measure, DM , is calculated. This is defined as

$$DM = \int_{Pulsar}^{Earth} n_e dl, \quad (1)$$

in which n_e is the electron density in the interstellar medium and dl considers the distance from the Pulsar to the Earth [2, p. 86]. The time delay in pulse arrival time, t , as a function of the pulse frequency, ν , is given by the equation:

$$t = \frac{DM}{2.410 \times 10^{-4} \nu_{MHz}^2}. \quad (2)$$

By plotting t versus ν , the gradient of this relationship will reveal the dispersion measure.

2.2. Timing the Crab Pulsar

With the knowledge of the dispersion measure, the time of arrival of the pulse can be obtained. A Time of

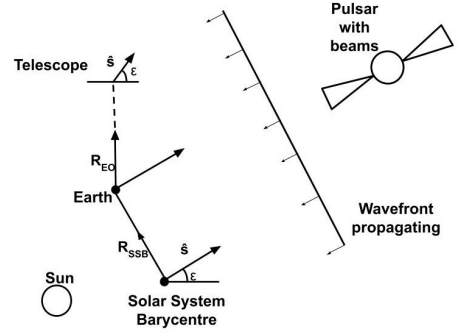


FIG. 1. This is a simplified diagram of the geometry required to perform a barycentric correction. The sun, Earth, SSB and the pulsar is depicted with relevant vectors and angles labelled. The diagram is not to scale.

Arrival estimation procedure [3] is followed to acquire the exact time that the pulse reaches the observatory with uncertainties. However, a barycentric correction must be taken to remove the effects of the Earth's rotation and orbit around the sun to ensure a consistent inertial frame. This is done by converting the pulse arrival time at the observatory to that of the solar system barycentre (SSB, the centre of mass of the solar system). The correction is depicted by Figure 1.

First, the SSB coordinates must be interpolated because they are published only once per day. Lagrange interpolation is therefore used to obtain the coordinates at the exact times of the pulse arrivals. This method constructs a polynomial to return a continuous function, allowing the determination of the SSB coordinates during the pulse time of arrival [4]. Next, delays of the pulse arrivals due to the motion of the Earth are considered. The Roemer delay, $\Delta_{R\odot}$, considers the light travel time between the telescope and the SSB [2, p. 207], and is calculated by:

$$\Delta_{R\odot} = \frac{-1}{c} (\vec{r}_{SSB} + \vec{r}_{EO}) \cdot \hat{s}, \quad (3)$$

in which c is the speed of light, $\vec{r}_{SSB}$ is the vector from the Earth to the SSB, $\vec{r}_{EO}$ is the vector from the telescope location to the geocentre, and $\hat{s}$ is the unit vector from the Barycentre to the Pulsar. Conveniently, since the pulsar is at a large distance, the expanded second term can be approximated as:

$$\vec{r}_{EO} \cdot \hat{s} \approx \frac{r_{\oplus} \sin \epsilon}{c}. \quad (4)$$

In equation (4), $r_{\oplus}$ represents the radius of the earth, taken as 6,371 km, and ε is the angle of the pulsar above the horizon. Performing this calculation gives the Roemer delay, which can be added to the Time of Arrival values to obtain the time when the pulsar signal reaches the barycentre. Following this, a trivial estimation of the pulsar's number of rotations N , can be derived by

$$N = \frac{t - t_0}{P}, \quad (5)$$

where t is the Time of Arrival at the barycentre, t_0 is the reference epoch, and P is an estimate period, taken as 0.033848293 [5]. Despite N being intended to be an integer, any slight difference in the model period will result in a fractional, periodic remainder known as the residual. Plotting the residual against the time elapsed results in a linear fit, from which the gradient reveals an improved value of the period which will overwrite P . Furthermore, the period derivative, $\dot{P}$, can be calculated if the rotational phase, $\phi(t)$, is written as a Taylor series:

$$\phi(t) = \phi_0 + \nu_0(t - t_0) + \frac{1}{2}\dot{\nu}_0(t - t_0)^2 + \dots, \quad (6)$$

where ν_0 is the spin frequency and $\dot{\nu}_0$ is the spin frequency derivative. This allows the calculation of the period derivative via $\dot{P} = -P^2\dot{\nu}_0$. From this, the surface magnetic field strength, B_{surf} , can be calculated by

$$B_{\text{surf}} = 3.2 \times 10^{19} \sqrt{P\dot{P}}, \quad (7)$$

[2, p. 62] and the characteristic age, T_c , as

$$T_c = \frac{P}{2\dot{P}}. \quad (8)$$

It should be noted that Equation (8) assumes a birth period of 10 ms [2, p. 61].

3. RESULTS AND DISCUSSION

3.1. The Dedispersion of Crab Pulsar Data

Figure 2 and 3 will show the plotting of the pulse arrival time against the inverse square of frequency. A linear fit will show the proportionality, allowing the calculation of the DM from the gradient following the methods in Equation 2.

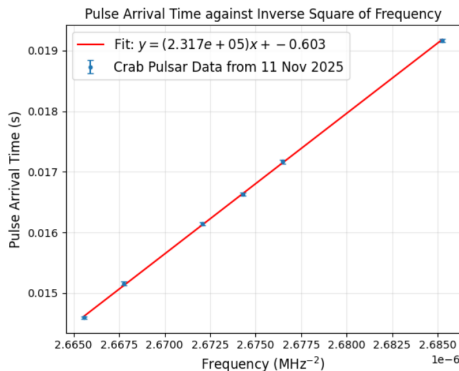


FIG. 2. This plot shows pulse arrival time versus inverse frequency squared for Crab Pulsar observations on November 11, 2025. A DM value of $(56.0 \pm 0.4) \text{ cm}^{-3}\text{pc}$ was obtained. A linear fit yields a reduced χ^2 of 1.01, indicating a good fit to the data.

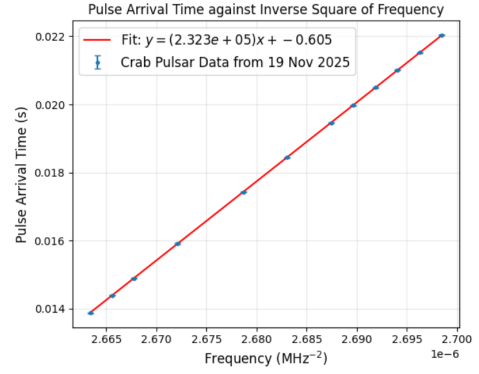


FIG. 3. This plot shows pulse arrival time versus inverse frequency squared for Crab Pulsar observations on November 19, 2025. A DM value of $(56.01 \pm 0.06) \text{ cm}^{-3}\text{pc}$ was obtained. A linear fit yields a reduced χ^2 of 0.34, suggesting slightly overestimated uncertainties.

Comparing the DM values from Figures 2 and Figure 3 with the known DM value of 56.7712 [5] shows a 2σ and 13σ discrepancy respectively. The former could have been due to random errors in various stages in measurement and data processing, such as ill-fitting a Gaussian on weaker signals. However, the latter suggests systematic errors, possibly from selective removal during Radio Frequency Interference (RFI) mitigation which introduced bias to the frequency bands used. Since the data required dedispersion prior to formulating the time of arrival of the pulse, the discrepancy in the DM affected the subsequent stages of this experiment.

3.2. Crab Pulsar Timing by Obtaining Residuals

Upon performing the barycentre corrections on the pulse arrival times, two linear fits were performed to obtain the pulse period and the period derivative.

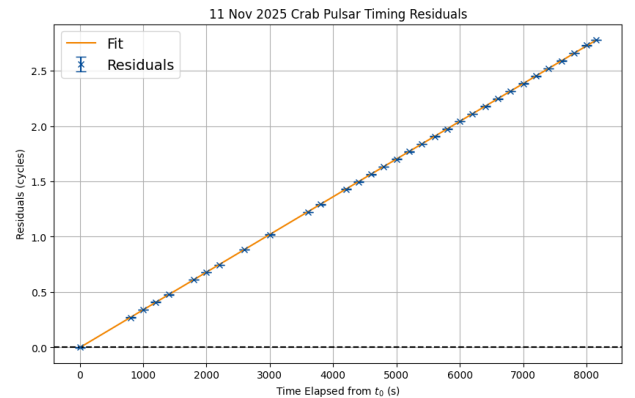


FIG. 4. This plot shows the residuals against the time elapsed from the reference epoch for the Crab Pulsar observations on November 11, 2025. An improved period of $(0.0338482160 \pm 1 \times 10^{-10}) \text{ s}$ was obtained. A linear fit yields a reduced χ^2 of 1.07, indicating a good fit.

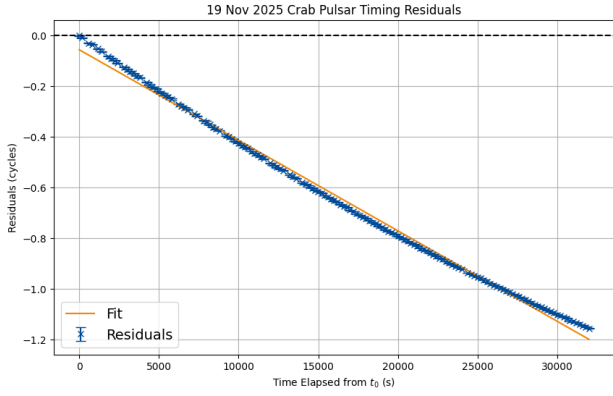


FIG. 5. This plot shows the residuals against the time elapsed from the reference epoch for the Crab Pulsar observations on November 19, 2025. An improved period of $(0.0338486878 \pm 2 \times 10^{-10})$ s was obtained. A linear fit yields a reduced χ^2 of 259, indicating a poor fitting.

Figures 4 and 5 show the fitting of measurements taken a week apart, revealing from the gradient that the pulsar's rotation speed has reduced as expected. Despite following the same analysis method fitting in Figure 5 was poor even though it was good for Figure 4, hinting towards unaccounted systematic errors. To investigate this, another iteration was performed and the residuals were fitted against the quadratic term of the Taylor expansion to obtain the period derivative.

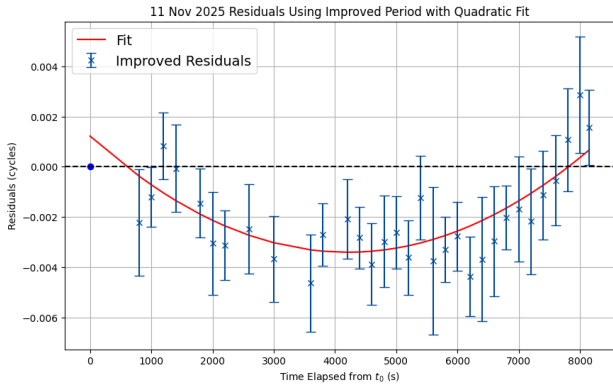


FIG. 6. This plot shows the improved residuals against the time elapsed from the reference epoch for the Crab Pulsar observations on November 11, 2025. A quadratic fit yields a reduced χ^2 of 0.41. The reference epoch has been removed.

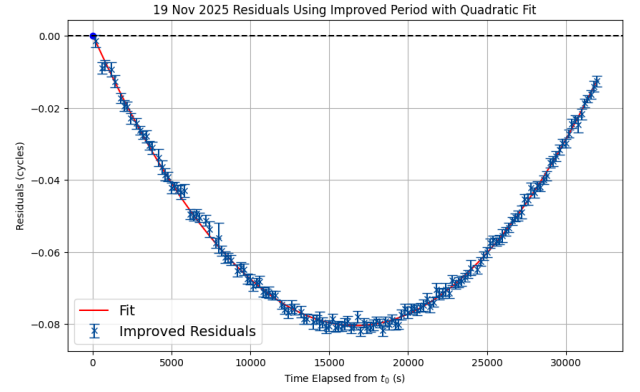


FIG. 7. This plot shows the improved residuals against the time elapsed from the reference epoch for the Crab Pulsar observations on November 19, 2025. A quadratic fit yields a reduced χ^2 of 0.55. The reference epoch has been removed.

The period derivatives of $(5.99 \pm 0.09) \times 10^{-13}$ and $(6.66 \pm 0.03) \times 10^{-13}$ were obtained from the quadratic fits in Figures 6 and 7, respectively. These values correspond to a surface magnetic field strength of $(4.6 \pm 0.33) \times 10^{12}$ G and a characteristic age of (827 ± 13.4) years. However, the referenced period value is 4.209×10^{-13} , with a surface magnetic field strength of $(4.6 \pm 0.33) \times 10^{12}$ G and a characteristic age of (827 ± 13.4) years [5]. Since some of the derived values deviated by more than 20 standard deviations from known values, the experimental results were deemed inconsistent. This inconsistency may have arisen from previous discrepancies, which could have been mitigated by observing the pulsar across more days for a better comparison. Furthermore, the limited observation time and the Earth's seasonal position arguably constrained the data quality.

4. CONCLUSION

The Crab Pulsar's period was obtained to be $(0.0338482160 \pm 1 \times 10^{-10})$ s and $(0.0338486878 \pm 2 \times 10^{-10})$ s across a one week period. While the measured dispersion measure and periodicity are within the same order of magnitude as published results, the standard deviations of the measured values indicate a significant inconsistency. The limitations that could have caused this were discussed throughout.

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